Exp. No. 13 Implement your own ping program.

Aim:-
To implement a ping program using python raw sockets to checks the reachability of a host and calculate the round trip time.

Code:-

```
import socket
import struct
import time
import os
def checksum(data):
    total=0
    for i in range(0, len(data)-1, 2):
        total += (data[i] << 8) + data[i+1]
    if len(data) % 2:
        total += data[-1] << 8
    total = (total >> 16) + (total & 0xffff)
    total += total >> 16
    return (~total) & 0xffff
def ping(host)
try:
    dest_addr = socket.gethostbyname(host)
    sock = socket.socket(socket.AF_INET,
                         socket.SOCK_RAW,
                         socket.IPPROTO_ICMP)
    packet_id = os.getpid() & 0xffff
    sequence = 1
    header = struct.pack(
        "!BBHHH", 8,
```

Sample Input/output:-

Enter destination host: 8.8.8.8

Reply from 8.8.8.8 : time = 24.53 ms

Result:-

Thus, the ping program was successfully implemented in python using raw sockets.

```
8,
0,
0, packet_id, sequence)
data = b"PING"
checksum = checksum(header + data)
header = struct.pack(
    "!BBHHH",
    8,
    0,
    checksum,
    packet_id,
    sequence )
packet = header + data
start = time.time()
sock.sendto(packet, dest_addr,))
sock.settimeout(2)
sock.recvfrom(1024)
end = time.time()
rtt = (end - start) * 1000
print("Reply from", dest_addr, ": time=", 
sock.close()        2)ms" %(rtt)
except socket.timeout:
    print("Request timed out")
except permission Error:
    print("Permission denied. Run the program
as administrator (root.")
host = input("Enter destination host: ")
ping(host)
```